Sales Data Analysis Using SQL

Introduction

This project presents a comprehensive analysis of an e-commerce transactional dataset using MySQL. The primary objective of the analysis is to transform raw transactional data into meaningful business insights through structured SQL queries and analytical techniques.

The project explores several key business areas, including sales performance, customer purchasing behavior, product performance, payment methods, coupon usage, referral sources, and operational order outcomes. Data cleaning and preprocessing techniques were also applied to improve data consistency and analytical accuracy.

Using SQL operations such as filtering, grouping, aggregation, sorting, and trend analysis, the project demonstrates how data can support business decision-making and performance evaluation. The analysis highlights the practical application of SQL in real-world business intelligence and data analytics workflows.

- Data cleaning
    1. Checked for null values and found 309 null values in the coupon code column
    2. These null values were replaced by no coupon with the assumption that the null values are for those who didn't use a coupon
    3. Checked for duplicated rows and found non
    4. Checked if there are prices that are negative or zero and found all prices to be positive
- Analysis

The analysis showed first that we have 1189 customers that performed 1200 orders these orders and there are as follows:

Table 1 number of customers and orders and average cart size

Average cart size	5.4850
Number of customers	1189
Number of orders	1200

Table 2 table that shows orders by year

2023	510
2024	459
2025	231

Table 3 table of total revenue per year only for successful delivered orders

Year	Revenue
2023	552643.24
2024	480235.87
2025	231882.85

Table 4 table that whose price of lowest and highest order

Highest vs lowest	Total price
Highest order	3456.40

Lowest order	11.39

Table 5 table that counts each order status category

Order Status	count
Cancelled	250
Returned	247
Pending	237
Shipped	235
Delivered	231

Our analysis reported that total revenue is 242600.32 and an average revenue per order of 1050.22 taking into consideration that only successful orders that were delivered

Table 6 revenue by payment method only for successful delivered orders

Payment Method	Revenue
Online	67119.65
Credit Card	46054.96
Debit Card	45885.14
Cash	42857.69
Gift Card	40682.88

Table 7 revenue by referral source for delivered orders

Referral Source	Revenue
Email	64553.52
Facebook	52149.38
Instagram	51299.56
Google	41406.25
Referral	33191.61

Table 8 revenue for coupon and non-coupon payments for delivered orders only

Revenue by coupon payments	Revenue by non-coupon payments
189126.24	53474.08

Table 9 Quantity delivered for each product

Product	Quantity delivered
Laptop	121
Chair	105
Phone	101
Monitor	96
Desk	94
Printer	88
Tablet	85

Table 10 revenue by delivered products

Product	Revenue
Laptop	40714.43
Phone	40345.41
Printer	38054.73
Monitor	35999.62
Tablet	31794.52
Chair	31465.83
Desk	24225.78

Table 11 maximum price for each product

Product	Maximum unit price
Laptop	699.93
Tablet	699.88
Desk	697.93
Printer	697.19
Monitor	696.71
Chair	695.29
Phone	695.10

Table 12 table with our top 10 customers

Customer ID	Total spent
C57276	3456.40
C67260	3390.80
C47778	3334.00
C53464	3299.25
C88029	3277.75
C89979	3170.00
C11415	2876.20
C50415	2830.35
C43595	2807.40
C13366	2714.20

Table 13 payment method frequency

Payment Method	count
Online	258
Cash	246
Credit Card	234
Debit Card	232
Gift Card	230

Table 14 Revenue from delivered by payment method

Payment Method	Revenue
Online	67119.65
Credit Card	46054.96
Debit Card	45885.14
Cash	42857.69
Gift Card	40682.88

Table 15 Average revenue by payment method for delivered orders

Payment Method	Average revenue by payment method
Gift Card	1099.54
Credit Card	1071.05
Debit Card	1067.10
Online	1065.39
Cash	952.39

Table 16 frequency of coupon usage

Coupon Code	Count
FREESHIP	313
No Coupon	309
WINTER15	292
SAVE10	286

Table 17 frequency of coupon usage for delivered orders

Coupon Code	Count
SAVE10	63
FREESHIP	61
WINTER15	55
No Coupon	52

Table 18 revenue by coupon code for delivered orders

Coupon Code	Revenue
SAVE10	69554.36
FREESHIP	65534.35
WINTER15	54037.53
No Coupon	53474.08

Table 19 average revenue from coupon payments and non-coupon paymnrts for delivered orders

Average revenue from coupon payments	Average revenue from non-coupon payments
1056.57	1028.35

Table 20 number of orders that came from every referral source

Referral Source	Orders
Instagram	259
Email	250
Google	241
Facebook	228
Referral	222

Table 21 number of delivered orders for every referral source

Referral Source	Orders
Email	60
Instagram	52
Google	44
Facebook	42
Referral	33

Table 22 revenue by referral source for delivered orders

Referral Source	Revenue
Email	64553.52
Facebook	52149.38
Instagram	51299.56
Google	41406.25
Referral	33191.61

Table 23 average revenue by referral source for delivered orders

Referral Source	Revenue
Facebook	1241.65
Email	1075.89
Referral	1005.81
Instagram	986.53
Google	941.05

Table 24 delivery, cancellation, return rates

Delivery rate	Return rate	Cancellation rate
19.25%	20.58%	20.88%

Table 25 products bought by coupons

Product	Orders
Printer	139
Tablet	133
Chair	130
Desk	129

Laptop	127
Phone	117
Monitor	116

Conclusion:

This project demonstrated how SQL can be used to transform raw sales transactional data into meaningful business insights. Through data cleaning, aggregation, filtering, grouping, and trend analysis using MySQL, the dataset was analyzed from multiple business perspectives including sales performance, customer behavior, product performance, coupon usage, referral effectiveness, payment methods, and operational order outcomes.

The analysis revealed that the business generated substantial revenue from successfully delivered orders, with laptops and phones contributing significantly to total sales revenue. Payment method analysis showed that online payments generated the highest revenue, while gift cards had the highest average revenue per delivered order. Referral source analysis indicated that email and social media channels played an important role in customer acquisition and revenue generation. Coupon analysis showed that promotional campaigns were widely used and contributed heavily to delivered-order revenue, with the SAVE10 coupon generating the highest revenue among all coupon codes. Customer-level analysis also identified a small group of high-value customers who contributed significantly to overall sales.

Operational analysis highlighted the importance of order fulfillment, as returned and canceled orders represented a noticeable portion of total transactions. By excluding returned transactions from revenue calculations, the analysis focused on realized business revenue and provided more accurate financial insights.

Overall, the project demonstrated practical SQL data analysis skills and emphasized the importance of analytical thinking in extracting actionable business insights from raw data.